

Supporting Evidence

Attachment Rescue · Track 1 (Ideas). All figures reproducible from public data with the linked code.

1. CENSUS OF THE FULL POPULATION (FILES LISTED BY NAME, NOTHING DOWNLOADED)

NCI-funded open-access papers, 2016–2022, with supplementary files and no data-repository accession	Count
Papers in population (Europe PMC, funder "NCI NIH HHS"; no text-mined accession for GEO, SRA, dbGaP, EGA, ArrayExpress, PRIDE, BioProject, MetaboLights, PDB or GenBank)	26,228
Papers whose full text was readable · attachments listed	25,138 · 73,795
Papers with at least one spreadsheet / CSV / tab-text attachment	4,759 (18.9%)
Spreadsheet / CSV / tab-text files (Excel alone 14,448; 15.9 GB)	15,458
Those papers licensed CC BY or CC0 (files may be re-deposited unchanged)	3,824 (80%)
Share of papers with a spreadsheet attachment, by year 2016 → 2022	14% · 17% · 18% · 20% · 22% · 20% · 19%

2. FOUR RANDOM SAMPLES OF 300 PAPERS, EVERY ATTACHMENT DOWNLOADED AND OPENED (124 PAPERS LATER FOUND TO CARRY A GEO ACCESSION ARE EXCLUDED)

Cancer type	Analysed	Any dataset*	Native spreadsheet	CC BY / CC0	Dataset files
Glioblastoma	271	53 (19.6%)	41 (15.1%)	41	106
Breast cancer	271	51 (18.8%)	41 (15.1%)	46	132
Colorectal cancer	273	53 (19.4%)	41 (15.0%)	47	112
Leukemia	261	45 (17.2%)	33 (12.6%)	33	112
Pooled	1,076	202 (18.8%)	156 (14.5%, CI 12–17%)	167 (83%)	462

*Dataset = largest table ≥ 50 rows $\times$ 3 columns, or $\geq 1,000$ rows; PDF/Word tables count only as numeric grids; peer-review files, reporting summaries and figures excluded. Largest table: 152,683 rows. Hand check of 105 machine decisions (verdicts published): 60/60 "dataset" calls were real data tables; every miss was a table under 50 rows. About half of rescued tables are primary measurements, half derived results. Filter check on 500 random population papers: 15% reference a repository somewhere; 13.5% of spreadsheet papers already have a Figshare identifier. Netting those out: roughly 2,800 to 3,900 papers, 8,000 to 11,000 buried dataset files.

3. TWO LIVE EXAMPLES A REVIEWER CAN OPEN

- **121-patient glioblastoma cohort table** (case, sex, age, location, IDH/TERT status, MGMT, ATRX) — *Acta Neuropathol Commun* 2018, PMC6193287, file 40478_2018_613_MOESM2_ESM.xlsx. CC BY. Listed in no catalogue.
- **118,273-row drug-screening dataset** — *Nat Commun* 2019, PMC6599391, file 41467_2019_10912_MOESM4_ESM.xlsx. CC BY. Listed in no catalogue.

4. DEPOSIT-READY PACKAGES

- 8 datasets from CC BY papers, unmodified, each with DataCite-style metadata.json (title, creators,

5. CODE, DATA, SITE (OPEN, MIT)

- Repository: github.com/OJ-IRO/attachment-rescue — census and classification code, raw census output, per-cancer results, packages.
- Public page with searchable list of the 462 dataset files and plain-language descriptions: attachment-rescue.vercel.app

6. PRIOR WORK CITED

- Gobeill et al., *Bioinformatics Advances* 2025 — 36.2 M PMC supplementary files extracted and indexed (PMC12371329).
- Federer et al., *PLOS ONE* 2018 (doi 10.1371/journal.pone.0194768); Colavizza et al., *PLOS ONE* 2020 (10.1371/journal.pone.0230416); Federer, 2022 (PMC9401135).

license, article DOI, PMC ID, funder, format, size, sheet names) and README. Not yet uploaded; see governance in Prompt 4.

- Ghosh et al., *JNCI* 2025 (PMC12782234); Boja et al., *JCO CC* 2026 (PMC13479720); Boyd et al., *JNCI* 2026 (10.1093/jnci/djag199).
- CBIIT INS-Data pipeline: github.com/CBIIT/INS-Data; INS disease-code mapper: github.com/CBIIT/ccdi-ins-icdo-mapping-llm.